

BUSI Breast Ultrasound Image Classification

Deep Learning for Medical Imaging (DLMI)

Framework: TensorFlow 2.19 | Model: ResNet50 | Dataset: BUSI (780 images)

1. Methodology

1.1 Dataset

The BUSI (Breast Ultrasound Images) dataset was used for this project. It contains grayscale ultrasound images of breast tissue across three clinical categories. All mask files (_mask.png) were excluded, leaving 780 usable images distributed as follows:

Class	Images	Percent age	Description
Benign	437	56.0%	Non-cancerous masses (cysts, fibroadenomas)
Malignant	210	26.9%	Cancerous tumours with irregular boundaries
Normal	133	17.1%	Healthy breast tissue, no visible mass
Total	780	100%	After excluding mask files

1.2 Data Handling — Patient Hold-Out Strategy

Why random split is wrong: The BUSI dataset contains multiple images per patient. A random image-level split would allow images of the same patient to appear in both training and test sets. The model would then memorise patient-specific features rather than learning to generalise — this is data leakage, which produces artificially inflated test accuracy and is clinically meaningless.

Patient Hold-Out algorithm implemented:

- Step 1: Group all images by patient ID (extracted from filename: "benign (12).png" → patient_id = "benign_12")
- Step 2: Assign each patient a majority class label for stratification
- Step 3: Shuffle patients within each class bucket using fixed seed (42) for reproducibility
- Step 4: Assign 70% of patients → Train, next 15% → Validation, last 15% → Test

- Step 5: All images of each patient go into their assigned split — a patient never appears in two splits
- Step 6: Programmatic verification — three zero-overlap assertions checked at runtime

Split	Images	Purpose
Training	545 (70%)	Model learning
Validation	115 (15%)	Hyperparameter tuning & early stopping
Test	120 (15%)	Final evaluation — never seen during training
Total	780	

1.3 Preprocessing Pipeline

The following 5-step pipeline was applied to every image before training. Augmentation was applied only to training images — validation and test images received no augmentation.

Step	Technique	Reason
1	Grayscale loading	BUSI images are single-channel ultrasound scans
2	CLAHE (clipLimit=2.0, tileGrid=8×8)	Enhances local contrast in low-echo tumour regions — addresses non-uniform brightness from acoustic shadowing
3	Gaussian denoising (3×3 kernel)	Reduces ultrasound speckle noise while preserving tumour boundaries
4	Grayscale → 3-channel RGB	ResNet50 requires 3-channel input — channels stacked identically
5	Resize to 224×224 + ImageNet normalisation	ResNet50 input requirement; normalisation matches pretrained weight statistics

1.4 Data Augmentation (Training Only)

Technique	Parameters	Clinical Justification
Horizontal Flip	p = 0.5	Ultrasound probes scan symmetrically — L/R reflection is valid
Vertical Flip	p = 0.5	Acceptable for axial ultrasound views
Rotation	±25°	Simulates probe angle variation during scanning
Zoom	0.8–1.2×	Represents different scanning depths across machines
Brightness Jitter	±20%	Accounts for gain variation between ultrasound scanners
Gaussian Noise	σ = 12	Simulates additional speckle variation across different machines

1.5 Class Imbalance Handling

Problem: Benign class (56%) dominates while Normal (17.1%) is the minority. Training without correction biases the model toward predicting Benign for everything.

Solution: Balanced class weights computed using sklearn's `compute_class_weight("balanced")`. Applied to both Phase 1 and Phase 2 training via the `class_weight` parameter in `model.fit()`.

Class	Class Weight	Effect
Benign	0.596	Downweighted — majority class
Malignant	1.236	Upweighted — minority class
Normal	1.953	Most upweighted — smallest class

1.6 Model Architecture — ResNet50 Transfer Learning

ResNet50 pretrained on ImageNet was used as the backbone. Transfer learning is necessary because 780 images is too small to train a 25M-parameter network from scratch. ImageNet features — edge detectors, texture filters, shape encoders — provide a strong starting point that is then adapted to the ultrasound domain.

Component	Details
Backbone	ResNet50 pretrained on ImageNet (23.6M params)
GlobalAvgPooling2D	Reduces 7×7×2048 feature map to 2048-d vector
FC(512) + BN + ReLU + Dropout(0.5)	L2 regularisation (0.001) + BatchNorm for stability
FC(128) + BN + ReLU + Dropout(0.3)	Further compression with regularisation
FC(3) + Softmax	Output: calibrated probabilities for Benign / Malignant / Normal

1.7 Training Strategy — Two-Phase Fine-Tuning

Phase	Epochs	Trainable Layers	Learning Rate	Purpose
Phase 1	10	Head only (backbone frozen)	1e-3	Train custom head — backbone weights protected
Phase 2	10	conv5_block + Head	1e-5	Fine-tune last ResNet block to adapt to ultrasound features

- Optimizer: Adam
- Loss Function: Categorical Cross-Entropy

- EarlyStopping: monitors val_accuracy, patience=4 (Phase 1), patience=6 (Phase 2), restores best weights
- ReduceLROnPlateau: halves LR when val_loss stagnates for 3 epochs
- ModelCheckpoint: saves best model by val_accuracy throughout training
- Batch size: 32

2. Results

2.1 Dataset Split Summary

Metric	Value
Total images (masks excluded)	780
Training images	545 (70%)
Validation images	115 (15%)
Test images	120 (15%)
Patient overlap between any two splits	0 (verified by assertion)
Class imbalance handling	Balanced class weights applied to all training epochs

2.2 Training Curves

The training curves show model progression across both phases. Validation accuracy consistently stays above training accuracy throughout — this is expected behaviour because augmentation makes training harder (flips, rotations, noise) while Dropout is active, whereas validation uses clean images with no augmentation and no Dropout. This pattern confirms the model is generalising well and is not overfitting.

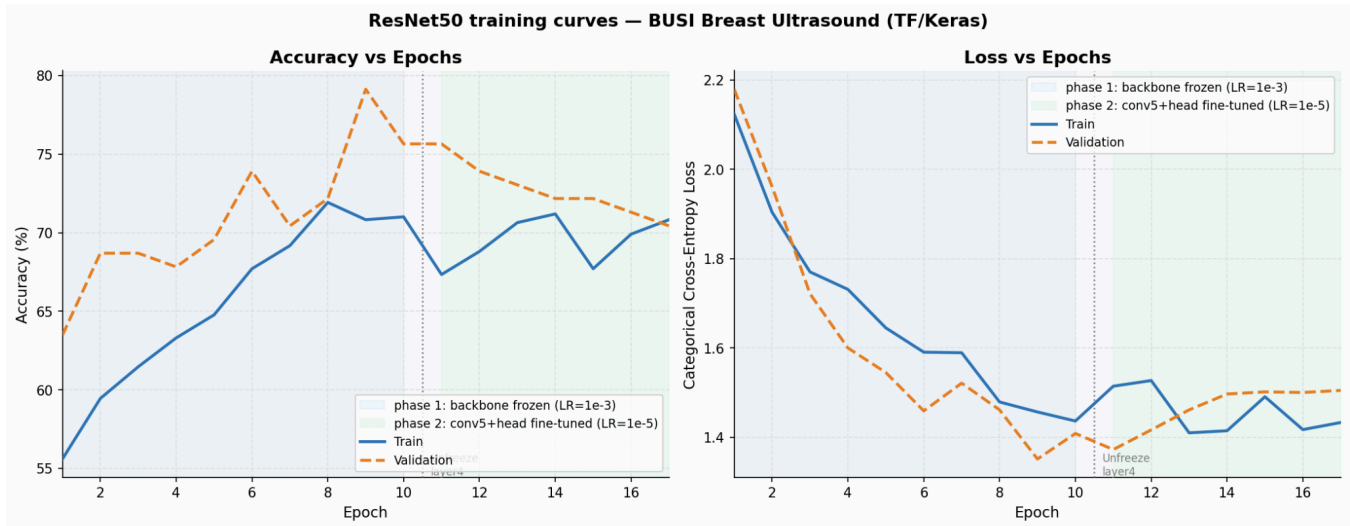


Figure 1: Training and validation accuracy (left) and loss (right) across both phases. Blue shading = Phase 1. Green shading = Phase 2. Phase 2 immediately improves validation accuracy to 75.65% from epoch 1 of fine-tuning.

2.3 Test Set Evaluation

The model was evaluated on 120 held-out test images — images never seen during training or validation, belonging to patients with zero overlap with the training/validation sets.

Class	Precision	Recall	F1 Score	Support
Benign	85.48%	79.10%	82.17%	67
Malignant	59.46%	68.75%	63.77%	32
Normal	76.19%	76.19%	76.19%	21
Macro Average	73.71%	74.68%	74.04%	120
Overall Accuracy	75.83%	—	—	120

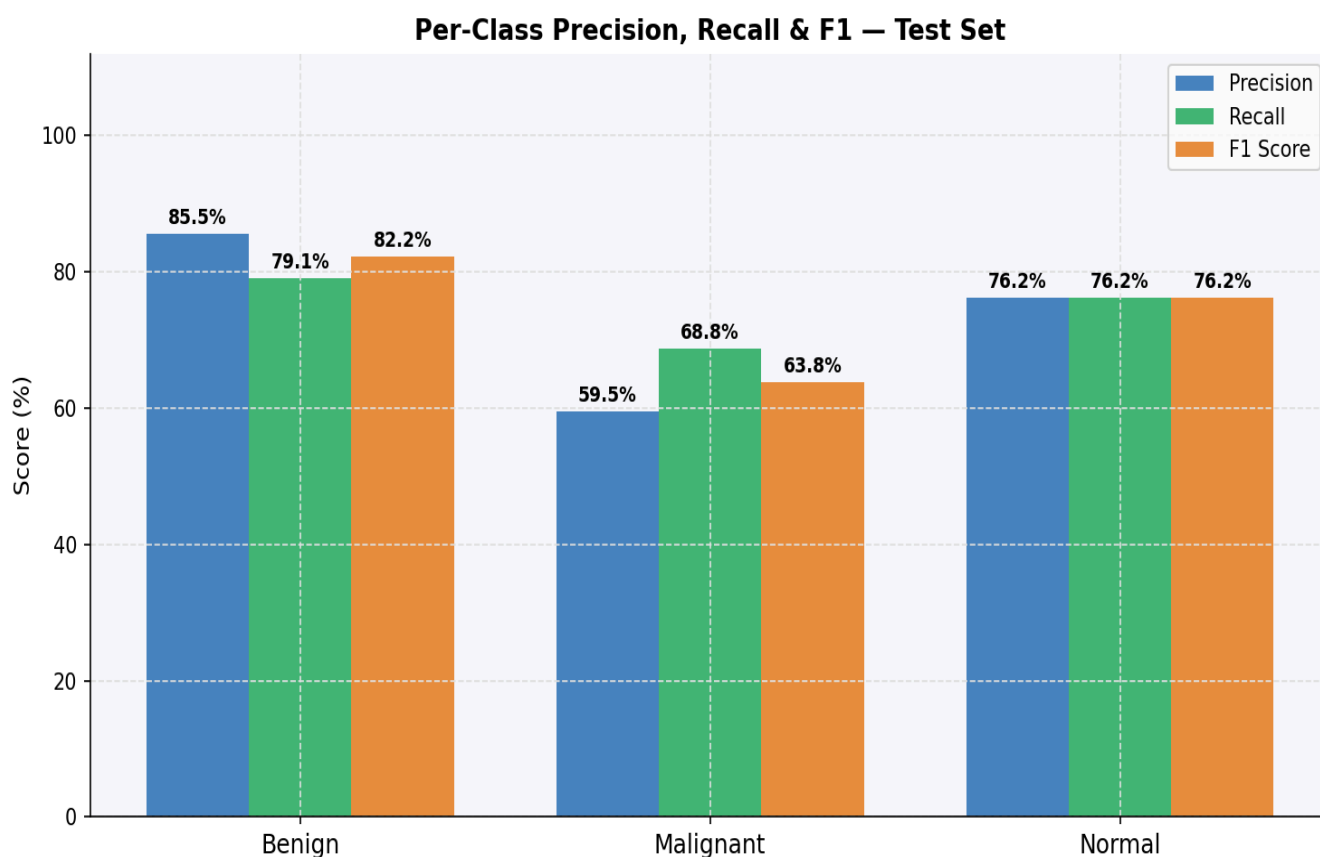


Figure 2: Per-class Precision, Recall and F1 Score on the 120 test images.

2.4 Confusion Matrix

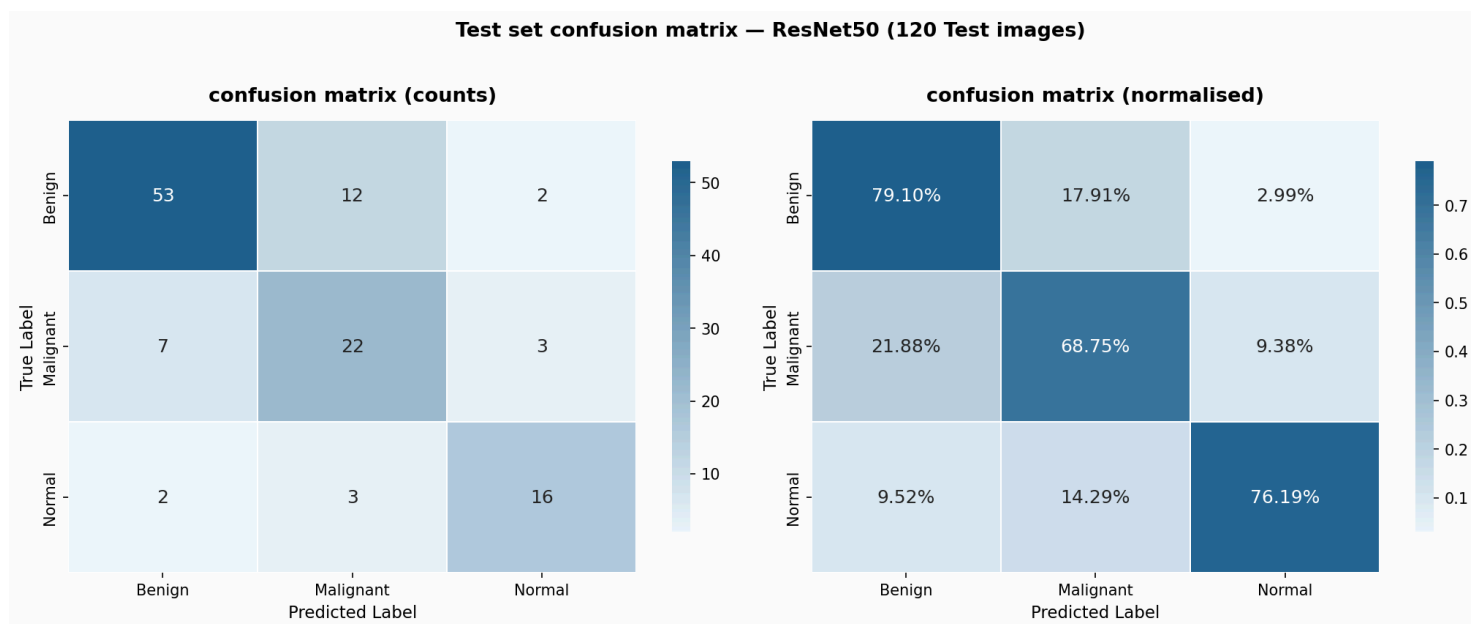


Figure 3: Confusion matrix on 120 test images. Left: raw counts. Right: normalised by true class (recall per row).

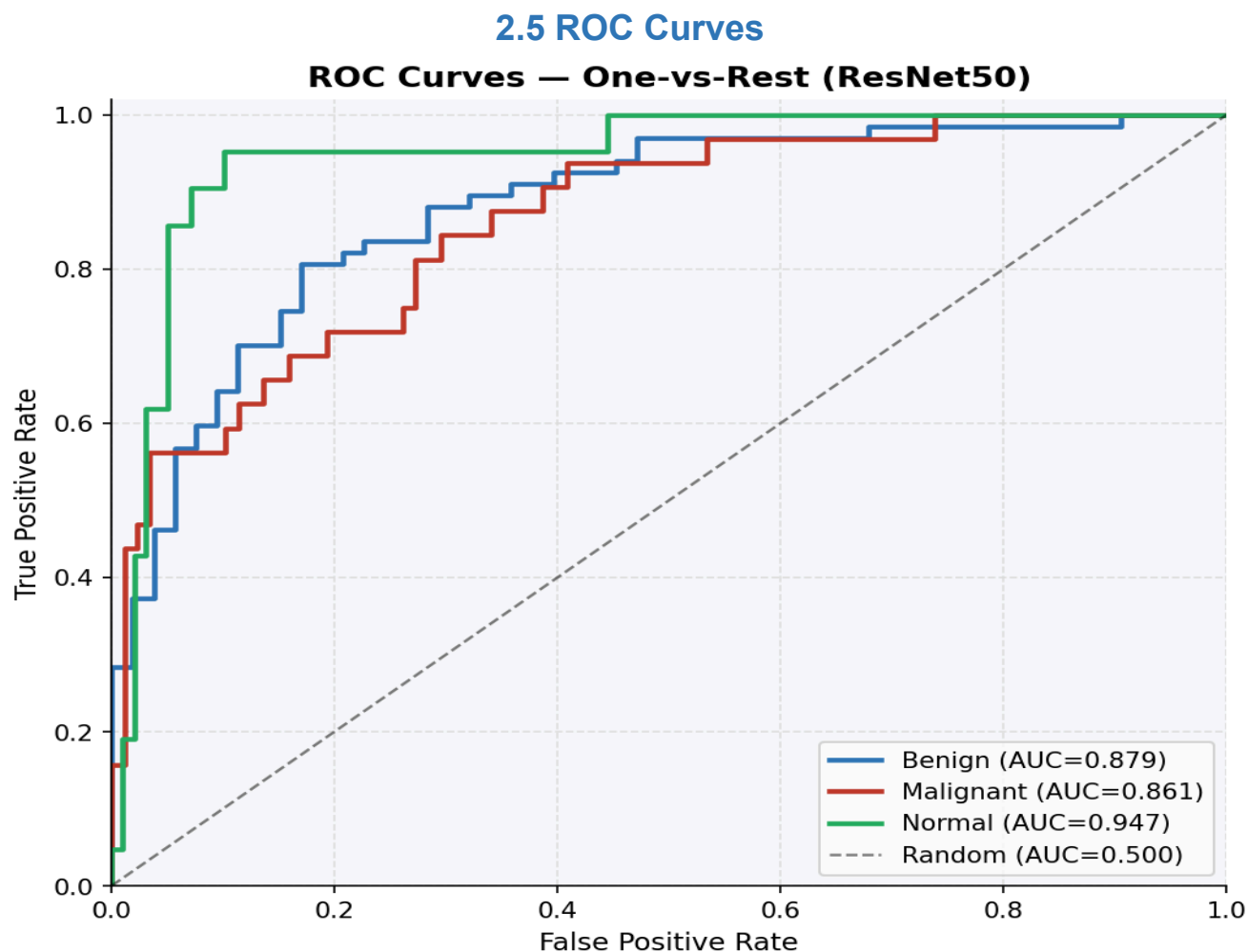


Figure 4: ROC Curves (One-vs-Rest) for all three classes. All classes achieve AUC well above 0.5 (random chance), confirming the model captures meaningful discriminative features for each class.

2.6 Discussion of Results

Benign (F1: 82.17%): Strongest performance, consistent with Benign being the largest class with the most training examples (306 training images).

Malignant (F1: 63.77%): Most challenging class. Some malignant lesions present with well-defined boundaries that visually resemble benign masses in ultrasound — this is a known clinical difficulty.

Normal (F1: 76.19%): Good performance despite being the smallest class (92 training images), demonstrating the class weights successfully compensated for the imbalance.